



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE (Advanced)

UNIT TEST

30-05-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TEAS)

ANSWER KEY

PAPER-1

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	D	B	C	D		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	A,B,C,D	A,B,C,D	A,B	A,C	B,C	B,D
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	8.00	38.00	26.67	12.50	1.33	4.00
SECTION-II (ii)	Q.	7	8	9			
	A.	7	3	6			

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4		
	A.	C	B	B	D		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	A,B,C,D	A,B,D	B,D	A	A,B,C,D	A,B,D
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	3.00	3.00	91.00	140.00	6.00	2.00
SECTION-II (ii)	Q.	7	8	9			
	A.	5	4	5			

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	B	A	D	D		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	B,D	A,C	A,C	C,D	A,C,D	A,B,C,D
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	5.00	3.00	125.00	385.00	3.33	17.00
SECTION-II (ii)	Q.	7	8	9			
	A.	6	6	7			

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (D)

$$l = \frac{m\ell^2}{12} = 20 \times \frac{3}{4} \times \frac{3}{4} \times \frac{1}{12} = \frac{180}{192}$$

2. Ans (B)

$$P = \frac{V^2}{4R} \Rightarrow R = \frac{V^2}{4P} = \frac{48 \times 48}{4 \times 200} = 2.88\Omega$$

3. Ans (C)

$$K_{eq1} = 2(8k \cos^2 30^\circ) + 2(8k \cos^2 60^\circ) \\ = 12k + 4k = 16k$$

$$\text{So, } K_{eq} = \frac{(16.16)}{32} k = 8k$$

4. Ans (D)

$$\frac{10}{100} \times 40 = 5 - \frac{5}{R+1} \times 1 \Rightarrow R = 4\Omega$$

PART-1 : PHYSICS

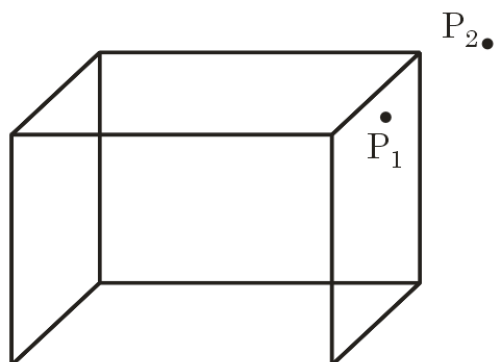
SECTION-I (ii)

5. **Ans (A,B,C,D)**

Using symmetry if charged particle lies at P_1

then $\phi_{OCDE} = \frac{q}{24\epsilon_0}$ and

$$\phi_{OABC} = \frac{\left(\frac{q}{\epsilon_0} - \frac{q}{8\epsilon_0}\right)}{3} = \frac{7q}{24\epsilon_0}$$

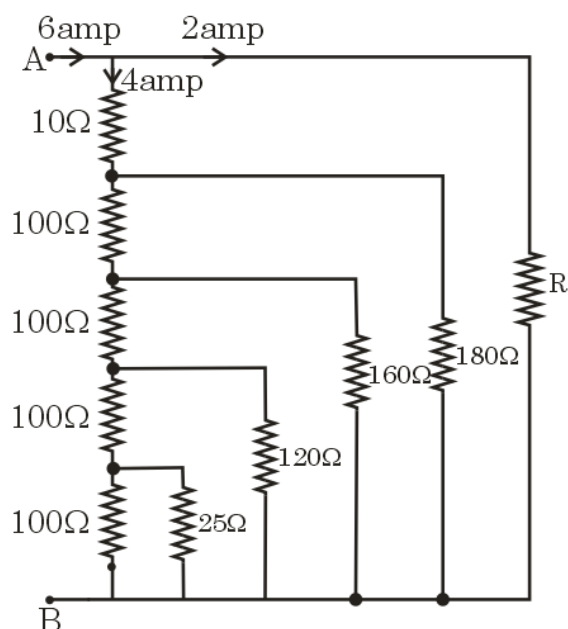


If the charge particle lies at P_2 then

$$\phi_{OCDE} = \frac{q}{24\epsilon_0}$$

$$\text{and } \phi_{OABC} = \frac{q}{24\epsilon_0}$$

6. **Ans (A,B,C,D)**



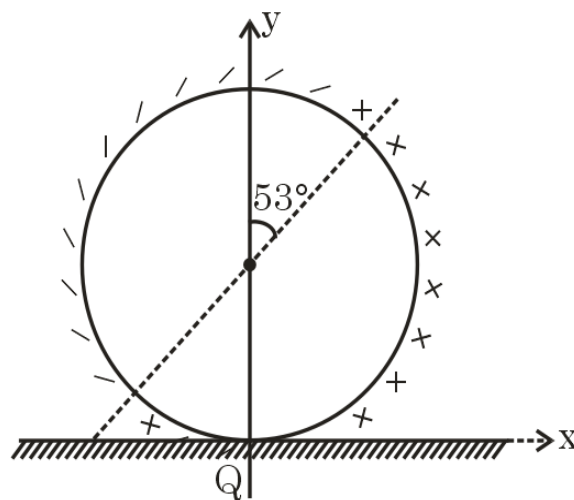
Equivalent resistance between P and Q = 100Ω .

$$\text{So current towards P} = \frac{400}{100} = 4 \text{ ampere}$$

Current through R = $6 - 4 = 2$ ampere

$$\text{So, } R = \frac{400}{2} = 200\Omega$$

7. **Ans (A,B)**



$$\tau_Q = P E \sin 37^\circ$$

$$= (\sigma 2\pi R^2) (R) E_0 \frac{3}{5}$$

$$= \frac{6}{5} \sigma \pi R^3 E_0 = \left(\frac{5}{3} m R^2 \alpha\right)$$

$$\alpha = \frac{18}{25} \frac{\sigma \pi R E_0}{m}$$

8. **Ans (A,C)**

$$\text{Angular velocity of earth } \omega_1 = \frac{\pi}{12} \text{ rad/hr}$$

$$\text{Angular velocity of satellite } \omega_2 = \frac{\pi}{3} \text{ rad/hr}$$

$$(\omega_2 - \omega_1)t = 2\pi$$

$$t = 8 \text{ hours}$$

→ for option 'B'

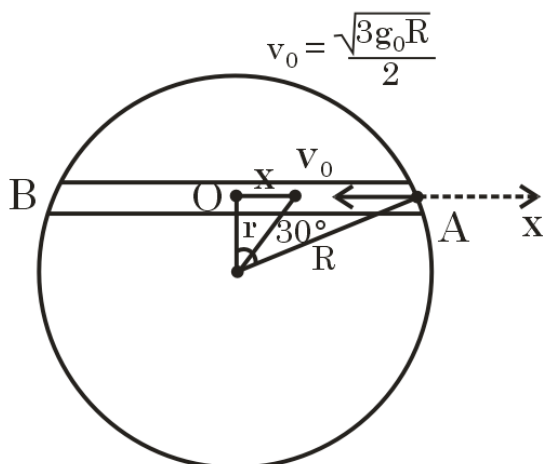
$$T^2 \propto r^3$$

$$T' = 48 \text{ hours}$$

$$\omega' = \frac{\pi}{24} \text{ hours}$$

$$t = 48 \text{ hours}$$

9. Ans (B,C)



$$\frac{d^2x}{dt^2} = -\frac{GMrcos\theta}{R^3} = -\left(\frac{GM}{R^3}\right)x = -\left(\frac{g_0}{R}\right)x$$

$$\omega = \sqrt{\frac{g_0}{R}}$$

$$\text{Now, } x = A \sin(\omega t + \alpha)$$

$$\text{at } t = 0, x = +R/2$$

$$\frac{R}{2} = A \sin \alpha \dots\dots(i)$$

$$V_x = \omega A \cos(\omega t + \alpha)$$

$$\therefore X = R \sin\left(\sqrt{\frac{g_0}{R}}t + \frac{5\pi}{6}\right)$$

$$-\frac{R}{2} = R \sin\left(\sqrt{\frac{g_0}{R}}t + \frac{5\pi}{6}\right)$$

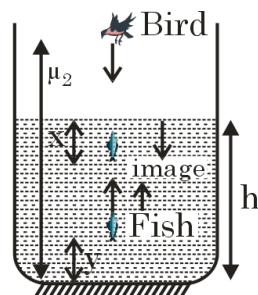
$$\frac{7\pi}{6} = \sqrt{\frac{g_0}{R}}t + \frac{5\pi}{6}$$

$$\therefore t = \frac{\pi}{3} \sqrt{\frac{R}{g_0}}$$

$$V_{\max} = \omega A$$

$$V_{\max} = \sqrt{\frac{g_0}{R}}R = \sqrt{g_0 R}$$

10. Ans (B,D)



Distance of image of fish as seen by bird only after refraction is

$$Z = (x - h) + (h - y) \frac{\mu_2}{\mu_1}$$

$$Z = (x - h) + (h - y) \frac{1}{4/3}$$

Differentiate w.r.t. time

$$\frac{dZ}{dt} = (-6 + 2) + (-2 - 4) \frac{3}{4} = -\frac{17}{2} \text{ cm/sec}$$

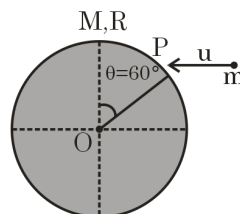
Similarly we can calculate for 2nd case.

PART-1 : PHYSICS

SECTION-II (i)

2. Ans (38.00)

Using conservation of angular momentum of the system about a horizontal axis passing through centre O.



$$muR \cos \theta = \left(\frac{MR^2}{2} + mR^2\right) \omega_0$$

$$muR \cos 60^\circ = \left(\frac{M}{2} + m\right) R^2 \omega_0$$

$$0.5 \times 20 \times \frac{1}{2} = 2.5 \times \frac{1}{4} \times \omega_0$$

$$\Rightarrow \omega_0 = 8 \text{ rad/s}$$

Now, using conservation of energy of the system

$$mgR(1 - \cos \theta) = \frac{1}{2} \left(\frac{MR^2}{2} + mR^2\right) (\omega_0^2 - \omega^2)$$

$$mg \frac{R}{2} = \frac{1}{2} \left(\frac{M}{2} + m\right) R^2 (\omega_0^2 - \omega^2)$$

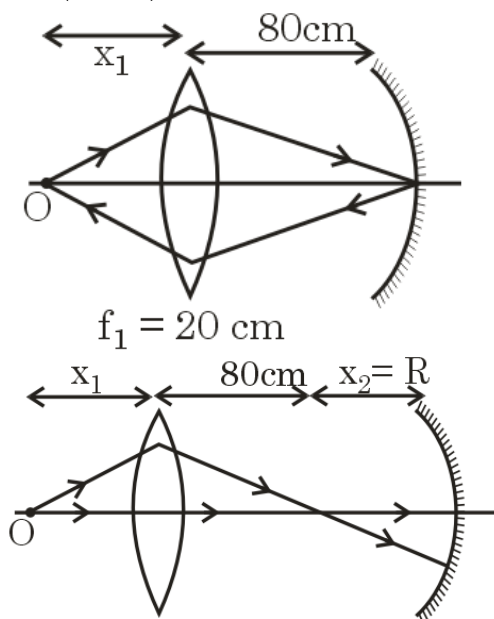
$$0.5 \times \frac{10}{2} = \frac{5}{4} \times \frac{1}{4} (\omega_0^2 - \omega^2)$$

$$\Rightarrow \omega_0^2 - \omega^2 = 8$$

$$\omega^2 = \omega_0^2 - 8 = 64 - 8 = 56$$

$$\omega = \sqrt{56} \text{ rad/s}$$

4. Ans (12.50)



$$\frac{1}{80} - \frac{1}{-x_1} = \frac{1}{20}$$

$$\frac{1}{x_1} = \frac{1}{20} - \frac{1}{80} \Rightarrow x_1 = 26.67 \text{ cm}$$

$$x_2 = R = 12.50 \text{ cm}$$

6. Ans (4.00)

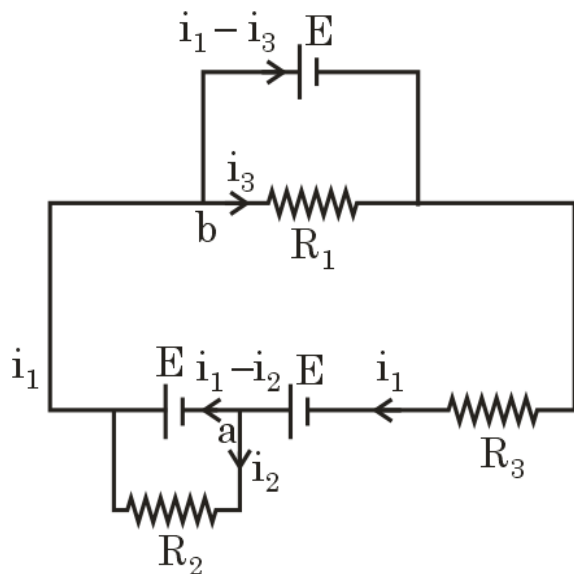
On solving by KVL we get,

$$i_1 = \frac{E}{R}, i_2 = \frac{E}{R}, i_3 = \frac{E}{R}$$

$$\Rightarrow i_1 - i_2 = \frac{2E}{R} \text{ and } i_1 - i_3 = 0$$

So reading of ammeter A_1 is $E/R = 4/3$

A_2 is $2E/R = 8/3$, A_3 is zero and $\Delta V_{ab} = 4V$



PART-1 : PHYSICS

SECTION-II (ii)

7. Ans (7)

$$\text{Speed of projection } (v_0) = v_0 = \sqrt{\frac{2GM}{R}} \dots\dots(1)$$

$$\frac{1}{2}mv^2 - \frac{GMm}{R} = \frac{1}{2}mv^2 - \frac{GMm}{r} \dots\dots(2)$$

$$v = \sqrt{\frac{2GM}{r}}$$

$$\int_R^{4R} \sqrt{r} dr = \sqrt{2GM} \int_0^t dt$$

$$t = \frac{7}{3} \sqrt{\frac{2R^3}{GM}}$$

8. Ans (3)

$$q_0 = \int \rho dv = \int_0^R \rho_0 r^n 4\pi r^2 dr$$

$$q = \int \rho dv = \int_0^{R/2} \rho_0 r^n 4\pi r^2 dr$$

$$\frac{kq_0}{R^2} = 16 \cdot \frac{kq}{\left(\frac{R}{2}\right)^2}$$

$$\text{So } n = 3$$

9. Ans (6)

Velocity of image :

$$8 \times \frac{2}{4/3} \times \frac{1}{2} = 6 \text{ cm/s}$$

PART-2 : CHEMISTRY

SECTION-I (i)

1. Ans (C)

Part of theory

2. Ans (B)

If FeO is like NaCl (rock salt)

Then no. of Fe^{2+} per unit cell will be = 4

crystal $\text{Fe}_{0.75}\text{O}$ contain some Fe^{2+} (x) & some

$\text{Fe}^{3+}(0.75-x)$ so $2x + (0.75 - x) \times 3 = 2$

$$\Rightarrow x = \frac{1}{4} \Rightarrow \text{number of } \text{Fe}^{3+} \text{ per unit cell} = 1 \times 2$$

PART-2 : CHEMISTRY

SECTION-I (ii)

5. **Ans (A,B,C,D)**

Correct order regarding atomic size :

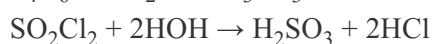
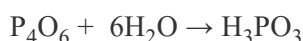
(A) $Zr \approx Hf$ (due to lanthanide contraction)

(B) $Fe \approx Co \approx Ni$ ($Z_{eff} \approx$ Screening effect)

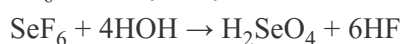
(C) $Sc < Y < La$ (Shells increasing)

(D) $Cu < Zn$ ($Z_{eff} <$ Screening effect)

6. **Ans (A,B,D)**



$SF_6 \rightarrow$ no hydrolysis



$\Rightarrow$ all are dibasic oxyacid

8. **Ans (A)**

If $T_1 = T_2 = T_3$ then molecular mass of gas 'A' > molecular mass of gas 'B' > molecular mass of gas 'C'.

9. **Ans (A,B,C,D)**

Moles of A = 10 moles

Moles of B = 40 moles

$$\therefore \text{Mole fraction} \Rightarrow x_A = \frac{10}{50}$$

$$x_B = \frac{40}{50}$$

$$\therefore P_T = P_A^0 x_A + P_B^0 x_B$$

$$\Rightarrow 100 \times \frac{10}{50} + 150 \times \frac{40}{50} = 140 \text{ mm of Hg}$$

$$\text{Mole fraction of 'A' in vapour phase} = \frac{100 \times \frac{10}{50}}{140} = \frac{1}{7}$$

$$\therefore \text{Mole fraction of 'B' in vapour phase} = \frac{6}{7}$$

PART-2 : CHEMISTRY

SECTION-II (i)

3. **Ans (91.00)**

$$Kt = \ln \left(\frac{V_\infty - V_0}{V_\infty - V_t} \right)$$

$$10^{-2} \times 230.3 = 2.303 \log \left(\frac{100 - 10}{100 - V_t} \right) \Rightarrow V_t = 91 \text{ ml}$$

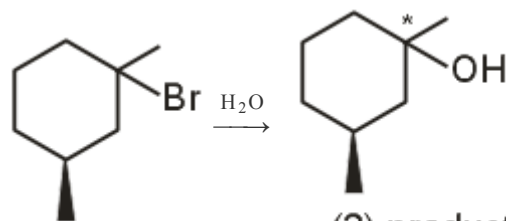
4. **Ans (140.00)**

$$r = K_1 [CH_3COO(CH_3)] \text{ where } K_1 = K [H^+][H_2O]$$

$$K_1 = (1.8 \times 10^{-3}) \times 0.1 \times \frac{1000}{18} = 10^{-2}$$

$$t = \frac{1}{K} \ln \frac{1}{0.25} = 140 \text{ sec}$$

6. **Ans (2.00)**



(2) products

PART-2 : CHEMISTRY

SECTION-II (ii)

8. **Ans (4)**

$$[\sigma(g)1s]^2, [\sigma(u)1s]^2, [\sigma(g)2s]^2, [\sigma^*(g)1s]^2, [\pi(u)2p_x]^1, [\pi(u)2p_y]^1$$

$$N_2 \Rightarrow [14 \text{ electrons}] = [\sigma(g)(1s)]^2, [\sigma(u)(1s)]^2, [\sigma(g)(2s)]^2, [\sigma^*(u)(2s)]^2, [\sigma(g)2p_z]^2, [\pi(u)2p_z]^2 = [\pi(u)2p_y]^2$$

$$C_2 \Rightarrow [12 \text{ electrons}] \Rightarrow [\sigma(g)(1s)]^2, [\sigma^*(u)(1s)]^2, [\sigma(g)(2s)]^2, [\sigma^*(u)(2s)]^2, [\pi(u)2p_x]^2 = [\pi(u)2p_y]^2$$

$$N_2^{2+} \Rightarrow [10 \text{ electrons}] [\sigma(g)(1s)]^2, [\sigma^*(u)(1s)]^2, [\sigma(g)(2s)]^2, [\sigma^*(u)(2s)]^2, [\pi(u)2p_x]^2 = [\pi(u)2p_y]^2$$

$$O_2 \Rightarrow [16 \text{ electrons}] \Rightarrow [\sigma(g)(1s)]^2, [\sigma^*(u)(1s)]^2, [\sigma(g)(2s)]^2, [\sigma^*(u)(2s)]^2, [\sigma(g)2p_z]^2, [\pi(u)(2p_x)]^2$$

$$= [\pi(u)(2p_y)]^2, [\pi^*(g)(2p_x)]^1 = [\pi^*(u)(2p_y)]^1$$

$$O_2^{2-} \Rightarrow [18 \text{ electrons}] [\sigma(g)(1s)]^2, [\sigma^*(u)(1s)]^2, [\sigma(g)(2s)]^2, [\sigma^*(u)(2s)]^2, [\sigma(g)2p_z]^2, [\pi(u)(2p_x)]^2$$

$$= [\pi(u)(2p_y)]^2, [\pi^*(g)(2p_x)]^2 = [\pi^*(u)(2p_y)]^2$$

PART-3 : MATHEMATICS

SECTION-I (i)

1. **Ans (B)**

$$y = x \sin x$$

$$\text{when } x = \frac{\pi}{2}$$

$$\Rightarrow y = \frac{\pi}{2}$$

$$\Rightarrow \text{curve passes through } p \left(\frac{\pi}{2}, \frac{\pi}{2} \right)$$

$$\frac{dy}{dx} = x \cos x + \sin x$$

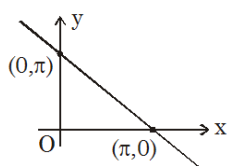
$$\left(\frac{dy}{dx} \right)_{\left(\frac{\pi}{2}, \frac{\pi}{2} \right)} = 1$$

$$\therefore \text{Slope of normal at } \left(\frac{\pi}{2}, \frac{\pi}{2} \right) = -1$$

$$\text{equation of normal is } y - \frac{\pi}{2} = -1 \left(x - \frac{\pi}{2} \right)$$

$$\text{i.e. } x + y = \pi$$

$$\therefore \text{Area of triangle} = \frac{1}{2} \cdot \pi \cdot \pi$$



$$A = \frac{\pi^2}{2}$$

$$\therefore [A] = 4$$

2. **Ans (A)**

$$\int_0^{\pi/2} \sin^4 x \, dx = \frac{3}{4} \times \frac{1}{2} \times \frac{\pi}{2} = \frac{3\pi}{16}$$

3. **Ans (D)**

$$\int_0^x \sqrt{1 - (f'(t))^2} \, dt = \int_0^x f(t) \, dt \quad 0 \leq x \leq 1$$

differentiating both the sides

$$\sqrt{1 - (f'(x))^2} = f(x)$$

$$\Rightarrow 1 - (f'(x))^2 = f^2(x)$$

$$\frac{f'(x)}{\sqrt{1 - f^2(x)}} = 1$$

$$\sin^{-1} f(x) = x + C$$

$$\therefore f(0) = 0 \Rightarrow C = 0 \Rightarrow f(x) = \sin x$$

$$\text{Now } \lim_{x \rightarrow 0} \frac{\int_0^x \sin t \, dt}{x^2} \left(\frac{0}{0} \right) = \frac{1}{2}$$

4. **Ans (D)**

$$I_1 = \frac{b-a}{2} = \frac{\pi/2 - 0}{2}$$

$$I_1 = \frac{\pi}{4}$$

$$I_2 = \int_{-\pi}^{\pi} \frac{\cos^2(x)}{1+a^x} \, dx$$

apply King and add

$$2I_2 = \int_{-\pi}^{\pi} \cos^2(x) \cdot dx = 2 \int_0^{\pi} \cos^2(x) \cdot dx$$

$$\Rightarrow I_2 = \int_0^{\pi} \cos^2 x \, dx = \frac{\pi}{2}$$

PART-3 : MATHEMATICS

SECTION-I (ii)

5. **Ans (B,D)**

$$\text{Curve is } (y-2)^2 = 2x^3 - 4 \quad \dots(1)$$

$$\frac{dy}{dx} = \frac{3x^2}{y-2}$$

$$\text{Eqn. of tangent at } (x_1, y_1) \text{ is } y - y_1 = \frac{3x_1^2}{y_1 - 2} (x - x_1)$$

As it passes through (1,2)

$$\Rightarrow 2 - y_1 = \frac{3x_1^2}{y_1 - 2} (1 - x_1) \Rightarrow (y_1 - 2)^2 = 3x_1^2 (x_1 - 1)$$

$$\Rightarrow 2x_1^3 - 4 = 3x_1^3 - 3x_1^2 \quad [\text{as } (y_1 - 2)^2 = 2x_1^3 - 4]$$

$$\Rightarrow x_1^3 - 3x_1^2 + 4 = 0 \Rightarrow (x_1 + 1)(x_1 - 2)^2 = 0 \Rightarrow x_1 = -1 \text{ or } 2$$

From eqn. (1) it is clear that $x_1 \neq -1$

$$\text{Thus } x_1 = 2 \Rightarrow y_1 = 2 \pm 2\sqrt{3}$$

Points of contact are $(2, 2 + 2\sqrt{3})$ and $(2, 2 - 2\sqrt{3})$

$\therefore$ sum of x-coordinates of points of contact is 4

$\therefore$ sum of y-coordinates of points of contact is 4.

6. Ans (A,C)

$F_1 : (2x - y - 9) + \lambda (3x + y - 21)$, F_1 passes through A (6, 3)

$F_2 : (2x + y - 5) + \mu (4x - y - 1) = 0$, F_2 passes through B (1, 3)

(A) maximum distance between parallel line of F_1 and F_2 is distance between their fix point $= \sqrt{5^2 + 0} = 5$

(C) If $L_1 \perp L_2$ then $m_{PB} \cdot m_{PA} = -1$

$$\begin{array}{c} P(h, k) \\ \swarrow \quad \searrow \\ A(6, 3) \quad (1, 3)B \end{array}$$

$$\Rightarrow \frac{k-3}{h-1} \cdot \frac{k-3}{h-6} = -1$$

$$\Rightarrow (h-1)(h-6) + (k-3)^2 = 0$$

$$\Rightarrow (x-1)(x-6) + (y-3)^2 = 0 \dots\dots (C)$$

7. Ans (A,C)

Put $x = \tan \theta$

$$f(x) = \sin^{-1} \left| \sin \frac{\theta}{2} \right| = \begin{cases} -\sin^{-1} \left(\sin \frac{\theta}{2} \right); & -\frac{\pi}{2} < \theta < 0 \\ \sin^{-1} \left(\sin \frac{\theta}{2} \right); & 0 \leq \theta < \frac{\pi}{2} \end{cases}$$

$$f(x) = \begin{cases} -\frac{1}{2} \tan^{-1} x; & -\infty < x < 0 \\ \frac{1}{2} \tan^{-1} x; & 0 \leq x < \infty \end{cases}$$

$$f'(x) = \begin{cases} \frac{-1}{2(1+x^2)}; & -\infty < x < 0 \\ \frac{1}{2(1+x^2)}; & 0 < x < \infty \end{cases}$$

$$\therefore f'(1) = \frac{1}{4}; f'(-1) = \frac{-1}{4} \Rightarrow \text{(A) is correct}$$

Range of $f(x) = \left[0, \frac{\pi}{4}\right)$. Hence (B) is incorrect.

Also $f'(x)$ is an odd function. $\Rightarrow$ (C) is correct

$$\lim_{x \rightarrow 0} \frac{f(x)}{x} \neq \frac{1}{2}$$

$$\left(\text{As, } f(0^+) = \frac{1}{2} \text{ and } f(0^-) = \frac{-1}{2} \right)$$

$\Rightarrow$ (D) is incorrect

8. Ans (C,D)

$$xy < 0 \Rightarrow x + \frac{1}{x} \geq 2, y + \frac{1}{y} \leq -2$$

$$\text{or } x + \frac{1}{x} \geq -2, y + \frac{1}{y} \geq 2$$

$$x + \frac{1}{x} \geq 2 \Rightarrow \sec^{-1} \left(x + \frac{1}{x} \right) \in \left[\frac{\pi}{3}, \frac{\pi}{2} \right)$$

$$y + \frac{1}{y} \leq -2 \Rightarrow \sec^{-1} \left(y + \frac{1}{y} \right) \in \left(\frac{\pi}{2}, \frac{2\pi}{3} \right] \Rightarrow z \in \left(\frac{5\pi}{6}, \frac{7\pi}{6} \right)$$

9. Ans (A,C,D)

$$f(x) = \begin{cases} 1 & x \in (2n\pi, 2n\pi + \frac{\pi}{2}) & n \in \mathbb{I}^+ \cup \{0\} \\ \frac{1+5\log_e x}{3} & x = (4n+1)\frac{\pi}{2} & n \in \mathbb{I}^+ \cup \{0\} \\ \frac{5\log_e x}{2} & x \in (2n\pi + \frac{\pi}{2}, 2n\pi + \frac{3\pi}{2}) & n \in \mathbb{I}^+ \cup \{0\} \\ \frac{1+5\log_e x}{3} & x = 2n\pi + \frac{3\pi}{2} & n \in \mathbb{I}^+ \cup \{0\} \\ 1 & x \in (2n\pi + \frac{3\pi}{2}, 2n\pi + 2\pi) & n \in \mathbb{I}^+ \cup \{0\} \end{cases}$$

$f(x)$ is discontinuous at all positive odd integral multiple of $\frac{\pi}{2}$

$f(x)$ is continuous at integral multiple of π

10. Ans (A,B,C,D)

$$\int \left(1 + \tan \frac{x}{2} \tan x \right) dx = \int \frac{\cos \frac{x}{2} \cos x + \sin \frac{x}{2} \sin x}{\cos x \cos \frac{x}{2}} dx$$

$$= \int \sec x dx = -\ln |\cos ex - \tan x| + C$$

$$\int \left(\tan x \cot \frac{x}{2} - 1 \right)$$

$$= \int \frac{\sin \frac{x}{2} dx}{\cos x \sin \frac{x}{2}}$$

$$= \int \sec x dx = \ln |\sec x + \tan x|$$

$$= \ln \left| \tan \left(\frac{\pi}{4} + \frac{x}{2} \right) \right| = -\ln \left| \frac{1 - \tan \frac{x}{2}}{1 + \cos \frac{x}{2}} \right| + C$$

PART-3 : MATHEMATICS

SECTION-II (i)

1. **Ans (5.00)**

$$\therefore \lim_{x \rightarrow \infty} \frac{f(x)}{|x|^3} = 0 \Rightarrow f(x) \text{ is either linear or quadratic}$$

$$\text{Let } f(x) = ax^2 + bx (\because f(0) = 0)$$

$$\therefore \lim_{x \rightarrow \infty} (\sqrt{f(x)} - x) = \lim_{x \rightarrow \infty} \frac{(a-1)x^2 + bx}{x\left(\sqrt{a + \frac{b}{x}} + 1\right)} = -1$$

$$\Rightarrow a - 1 = 0 \text{ and } \frac{b}{\sqrt{a} + 1} = -1 \Rightarrow b = -2$$

$$\text{Hence, } f(x) = x^2 - 2x$$

Now proceed.

2. **Ans (3.00)**

$$\therefore \lim_{x \rightarrow \infty} \frac{f(x)}{|x|^3} = 0 \Rightarrow f(x) \text{ is either linear or quadratic}$$

$$\text{Let } f(x) = ax^2 + bx (\because f(0) = 0)$$

$$\therefore \lim_{x \rightarrow \infty} (\sqrt{f(x)} - x) = \lim_{x \rightarrow \infty} \frac{(a-1)x^2 + bx}{x\left(\sqrt{a + \frac{b}{x}} + 1\right)} = -1$$

$$\Rightarrow a - 1 = 0 \text{ and } \frac{b}{\sqrt{a} + 1} = -1 \Rightarrow b = -2$$

$$\text{Hence, } f(x) = x^2 - 2x$$

Now proceed.

3. **Ans (125.00)**

$$f(n) = \frac{4}{\pi} \sum_{i=1}^n \sum_{j=1}^n \cot^{-1} \left(\frac{i}{j} \right)$$

replace i by j and j by i

$$f(n) = \frac{4}{\pi} \sum_{i=1}^n \sum_{j=1}^n \cot^{-1} \left(\frac{j}{i} \right)$$

add

$$2f(n) = \frac{4}{\pi} \sum_{i=1}^n \sum_{j=1}^n \frac{\pi}{n}$$

$$f(n) = n^2$$

$$f(5) + f(100) = 125$$

4. **Ans (385.00)**

$$f(n) = \frac{4}{\pi} \sum_{i=1}^n \sum_{j=1}^n \cot^{-1} \left(\frac{i}{j} \right)$$

replace i by j and j by i

$$f(n) = \frac{4}{\pi} \sum_{i=1}^n \sum_{j=1}^n \cot^{-1} \left(\frac{j}{i} \right)$$

add

$$2f(n) = \frac{4}{\pi} \sum_{i=1}^n \sum_{j=1}^n \frac{\pi}{n}$$

$$f(n) = n^2$$

$$\sum_{r=1}^{10} f(r) = \frac{10 \cdot 11 \cdot 21}{6} = 385$$

5. **Ans (3.33)**

$$\int_{-3}^1 f(x) dx = \int_0^1 f(x) dx + \int_0^3 f(x) dx$$

$$= \frac{1}{2} + \int_0^1 f(x) dx + \int_1^2 f(x) dx + \int_2^3 f(x) dx$$

$$= 1 + \left(\frac{x^3}{3} - x^2 + 2x \right)_1^2 + \frac{1}{2} \times 1 \times 2$$

$$\int_{-3}^1 f(x) dx = 1 + 1 + \frac{8}{3} - \left(\frac{1}{3} - 1 + 2 \right) = \frac{10}{3}$$

6. **Ans (17.00)**

$$\int_0^1 f(x) dx = \frac{1}{2}$$

$$\int_2^3 f(x) dx = 1$$

$$\int_1^2 f(x) dx = \frac{4}{3}$$

$$\frac{\int_0^1 f(x) dx + \int_2^3 f(x) dx}{\int_1^2 f(x) dx} = \frac{\frac{1}{2} + 1}{\frac{4}{3}}$$

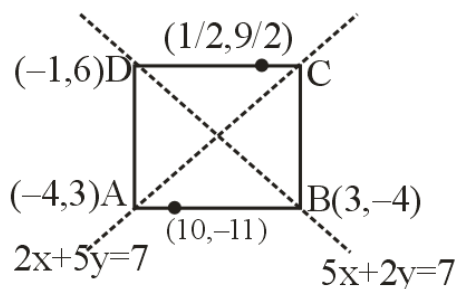
$$= \frac{3}{2} \times \frac{3}{4} = \frac{9}{8}$$

$$\therefore m + n = 9 + 8 = 17$$

PART-3 : MATHEMATICS

SECTION-II (ii)

7. Ans (6)



$$\frac{2x + 5y - 7}{\sqrt{29}} = \pm \frac{5x + 2y - 7}{\sqrt{29}}$$

$$\Rightarrow 3x - 3y = 0 \Rightarrow x - y = 0$$

$$7x + 7y = 14 \Rightarrow x + y = \frac{2}{7}$$

$$\text{So } AB : x + y = -1 \Rightarrow B(3, -4) \Rightarrow BC : x - y = 7$$

$$CD : x + y = 5 \Rightarrow D(-1, 6) \Rightarrow AD : x - y = -7$$

$$\Delta = \frac{6}{\sqrt{2}} \cdot \frac{14}{\sqrt{2}} = 42$$

8. Ans (6)

$$\begin{aligned} k &= \int_{-2022}^{2022} \frac{1}{(1 + x^{2021} + \sqrt{1 + x^{4042}})(1 + x^{2021} - \sqrt{1 + x^{4042}})} \cdot dx \\ &= \int_{-2022}^{2022} \frac{1 + x^{2021} - \sqrt{1 + x^{4042}}}{2 \cdot x^{2021}} \cdot dx \\ &= \int_{-2022}^{2022} \underbrace{\frac{1}{2 \cdot x^{2021}}}_{\text{odd function}} dx + \int_{-2022}^{2022} \frac{1}{2} \cdot dx - \int_{-2022}^{2022} \underbrace{\frac{\sqrt{1 + x^{4042}}}{x^{2021}}}_{\text{odd function}} dx \\ &= 2022 \end{aligned}$$

$\Rightarrow$ sum of digits is 6

9. Ans (7)

$$\begin{aligned} L &= \lim_{n \rightarrow \infty} \left(\frac{n+2}{n^2+n+1} + \frac{n+4}{n^2+2n+2^2} + \dots + \frac{n+2(2n)}{n^2+(2n)n+(2n)^2} \right) \\ &= \lim_{n \rightarrow \infty} \sum_{r=1}^{2n} \frac{n+2r}{n^2+r n+r^2} \\ &= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^{2n} \frac{1+2\left(\frac{r}{n}\right)}{\left(\frac{r}{n}\right)^2+\left(\frac{r}{n}\right)+1} \\ &= \int_0^2 \frac{1+2x}{x^2+x+1} dx = \ln(x^2+x+1) \Big|_0^2 = \ln 7 \\ e^L &= 7 \end{aligned}$$

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TEAS)

ANSWER KEY

PAPER-2

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	B	A	B	D	C	D	A	B	B
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	A	B	A	C	A	D	C	C	A	D
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	11	2	0	100	2	3	2	1500	5	2

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	C	A	D	C	D	C	D	C	B
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	D	A	B	B	B	C	C	B	D	A
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	6	22	3	3	64	1	2	2	5	6

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	B	A	C	B	D	C	D	C	A
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	A	B	D	C	B	D	C	C	D	A
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	5	0	1	15	1	6	4	3	5050	2

HINT – SHEET

PART-1 : PHYSICS

SECTION-I

1. **Ans (B)**

$$\text{Current in EC} = \frac{40}{20} = 2\text{A}$$

Current at C = $2 + 2 = 4\text{A}$

Equivalent resistance of CD = $\frac{5 \times 5}{10} = 2.5\Omega$

Potential drop in CD = $4 \times 2.5 = 10V$

$$100 - V_{EC} + V_{CD} + V_{DE}$$

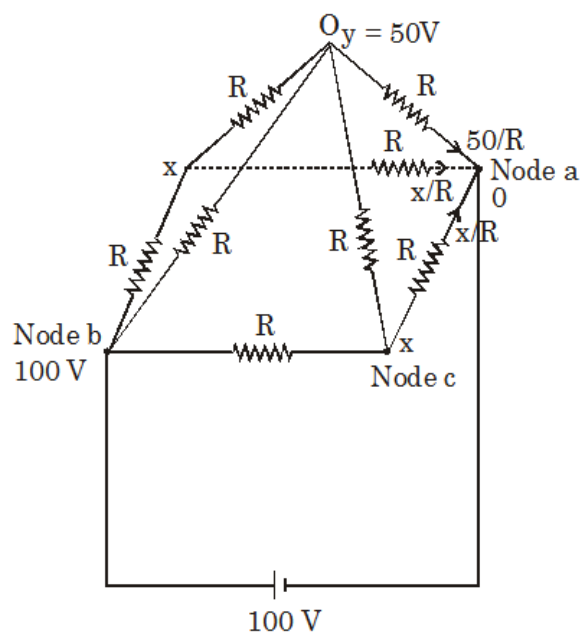
$$V_{DE} = 100 - 40 - 10 = 50V$$

Current in wire DG = 2A

Current in resistance DF = 2A

$$R \times 2 = 50 \text{ or } R = 25\Omega$$

2. Ans (B)



Step 2: By symmetry $100 - y = y - 0$ or $y = 50V$

Step 3: Now apply KCL at node c

$$\frac{x - 100}{R} + \frac{x + 50}{R} + \frac{x - 0}{R} = 0$$

or $x = 50V$

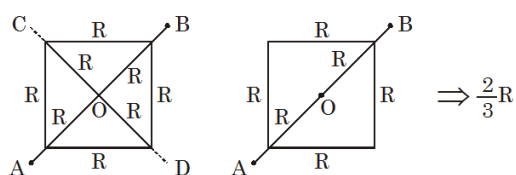
Step 4: If R_{eq} is equivalent resistance of this circuit, we have

$$\frac{100}{R_{eq}} = \frac{x}{R} + \frac{50}{R} + \frac{50}{12} \text{ or } R_{eq} = \frac{2}{3}R$$

Alternative method:

Making 2D projection

There is perpendicular axis of symmetry, i.e., the points COD will be equipotential then the circuit will be



3. **Ans (A)**

Resistance of each part = $\frac{R}{2n}$. For 'n' such parts connected in series, equivalent resistances, say

$$R_1 = n \left[\frac{R}{2n} \right] = \frac{R}{2}$$

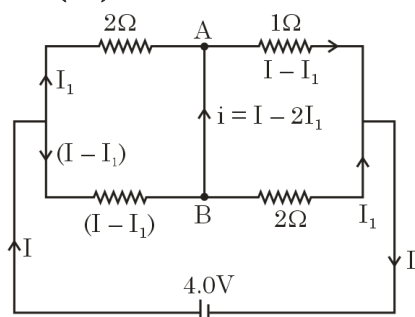
Similarly, equivalent resistance say R_2 for another set of n identical respectively in parallel

$$\text{would be } \frac{1}{n} \left(\frac{R}{2n} \right) = \frac{R}{2n^2}$$

For getting maximum of R_1 & R_2 , they should be connected in series & hence,

$$R_{eq} = R_1 + R_2 = \frac{R}{2} \left(1 + \frac{1}{n^2} \right)$$

4. **Ans (B)**



Using KVL, we get

$$2I_1 - (I - I_1) = 0 \text{ or } I = 3I_1$$

$$\text{and } (I - I_1) + 2I_1 = 4$$

Solving, we get $I_1 = 1A$ and $I = 3A$

Therefore, current through AB = $I - 2I_1$

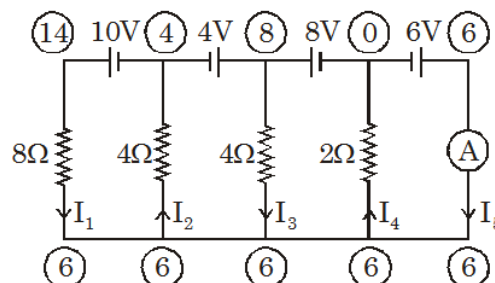
$$= (3 - 2)A = 1A$$

5. **Ans (D)**

$$k = \frac{E_1}{100} = \frac{2}{100} = 0.02 \text{ V cm}^{-1}$$

$$E_2 = k(75) = 0.02 \times 75 = 1.5 \text{ V}$$

6. **Ans (C)**



The current through ammeter

$$\begin{aligned} I &= I_4 - I_3 + I_2 - I_1 \\ &= \left(\frac{6-0}{2} \right) - \left(\frac{8-6}{4} \right) + \frac{(6-4)}{4} - \frac{(14-6)}{8} \\ &= 3 - \frac{1}{2} + \frac{1}{2} - 1 = 3 - 1 = 2A \end{aligned}$$

7. **Ans (D)**

Let ρ = density of sphere, R = radius of sphere, r = radius of the spherical cavities.

$$\text{Mass of the complete sphere} = \frac{4}{3}\pi R^3 \rho = M$$

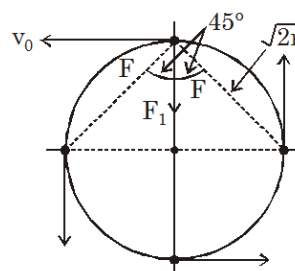
$$\text{Mass of the removed sphere} = \frac{4}{3}\pi r^3 \rho = m$$

$$\text{Here } m = \frac{Mr^3}{R^3} = \frac{M(1)^3}{(4)^3} = \frac{M}{64}$$

$$\text{Now } \vec{I}_R = \vec{I} + \vec{I}_P + \vec{I}_Q$$

$$\text{Here } I = 0, \text{ also } \vec{I}_P = -\vec{I}_Q \Rightarrow I_R = 0$$

8. **Ans (A)**



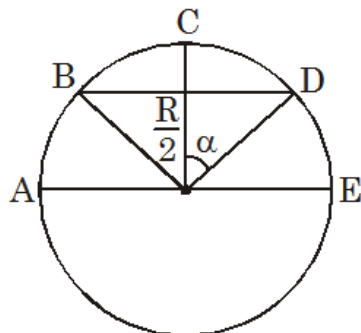
Centripetal force = net gravitational force

$$\Rightarrow mv_0^2 = 2F \cos 45^\circ + F_1 = \frac{2Gm^2}{(\sqrt{2})^2} \frac{1}{\sqrt{2}} + \frac{Gm^2}{4r^2}$$

$$\frac{mv_0^2}{r} = \frac{GM^2}{4r^2} [2\sqrt{2} + 1]$$

$$\Rightarrow v_0 = \left[\frac{Gm(2\sqrt{2} + 1)}{4r} \right]^{\frac{1}{2}}$$

9. Ans (B)



$$\alpha = 60^\circ$$

Solid angle subtended by BCD is

$$\omega = 2\pi(1 - \cos \alpha) = \pi$$

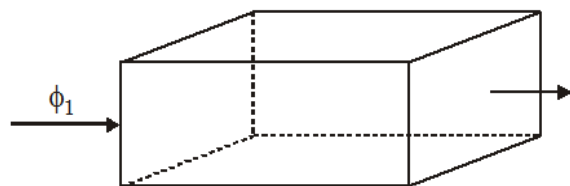
Solid angle subtended by ABDE is

$$\omega_{(ABCDE)} - \omega_{(BCD)} = 2\pi - \pi = \pi$$

Hence, flux through ABDE is

$$\phi = \frac{q}{\epsilon_0} \frac{\pi}{4\pi} = \frac{q}{4\epsilon_0}$$

10. Ans (B)



$$\phi_1 = -E_1 a^2 = -\alpha a^{1/2} a^2 = -\alpha a^{5/2}$$

$$\text{and } \phi_2 = E_2 a^2 = \alpha(2a)^{1/2} a^2 = \sqrt{2}\alpha a^{5/2}$$

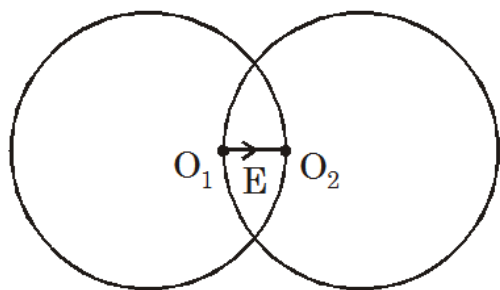
$$\phi_{\text{net}} = (\sqrt{2} - 1)\alpha a^{5/2}$$

Using the Gauss theorem, we get

$$\phi = \frac{q_{\text{in}}}{\epsilon_0}$$

$$\text{or } q_{\text{in}} = \phi \epsilon_0 = (\sqrt{2} - 1)\epsilon_0 \alpha a^{5/2}$$

11. Ans (A)



$$d = R$$

Electric field from O_1 through O_2 is uniform and is $E = \rho R / 3\epsilon_0$

So potential difference is

$$\Delta V = ER = \frac{\rho R^2}{3\epsilon_0} = \frac{\rho d^2}{3\epsilon_0}$$

13. Ans (A)

Charge can be considered as located at a distance R from the center. Total charge is

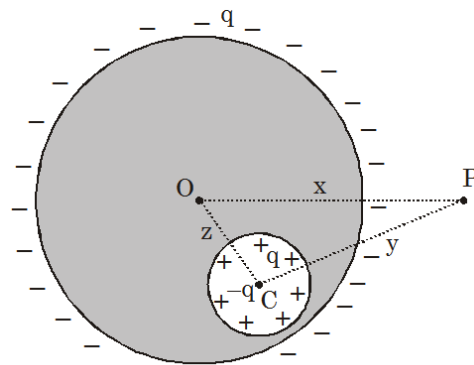
$$(Q - 2Q + 3Q) = 2Q$$

$$\text{Hence } V = \frac{1}{4\pi\epsilon_0} \frac{2Q}{R} = \frac{Q}{2\pi\epsilon_0 R}$$

17. Ans (C)

$$T = 2\pi \sqrt{\frac{\ell}{g \left(1 + \frac{\ell}{R}\right)}}$$

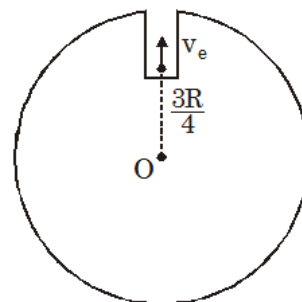
18. Ans (C)



$$V_P = \underbrace{(V_P)_{\text{cavity's induced charge}} + (V_P)_{-q}}_0 + (V_P)_{\text{surface charge}}$$

$$V_P = -\frac{kq}{x}$$

19. Ans (A)



COE

$$\frac{1}{2}mv_e^2 + mv = 0 \quad \dots (i)$$

$$v = \frac{-GM}{2R^3} (3R^2 - r^2)$$

$$v = \frac{-GM}{2R^3} \left(3R^2 - \frac{9R^2}{16} \right)$$

$$v = \frac{-39GM}{32R}$$

$$\text{by equation (i)} \quad \frac{1}{2}mv_e^2 + \left(\frac{-39GMm}{32R} \right) = 0$$

$$v_e = \sqrt{\frac{39Gm}{16R}} = \frac{1}{4} \sqrt{\frac{39Gm}{R}}$$

20. **Ans (D)**

For $\theta < \theta_C$ (critical angle), $\delta = r - i$

For $\theta > \theta_C$, $\delta = \pi - 2i$

PART-1 : PHYSICS

SECTION-II

9. **Ans (5)**

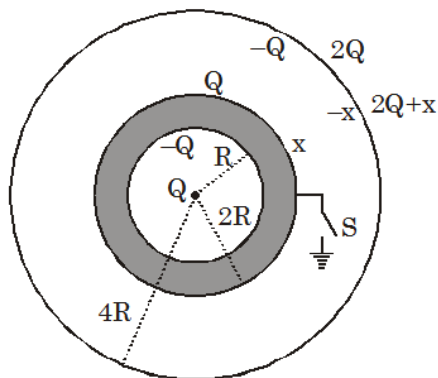
$$J = \sigma E$$

$$i/A = \sigma E \Rightarrow E = \frac{i}{\sigma A}$$

$$E \propto 1/A$$

$$\frac{E_1}{E_2} = \frac{A_2}{A_1} = \frac{10^2}{2^2} = 25$$

10. **Ans (2)**



$$v = 0$$

$$\frac{kx}{2R} + \frac{k2Q}{4R}$$

$$\frac{x}{2} = -\frac{2Q}{4R}$$

$$x = -Q$$

charge flown into the earth = $2Q$

PART-2 : CHEMISTRY

SECTION-I

5. **Ans (C)**

Fact

7. **Ans (C)**

$$p = P_A^\circ x_A + P_B^\circ x_B$$

$$600 = \left(\frac{3}{3+2}\right) P_A^\circ + P_B^\circ \left(\frac{2}{2+3}\right); 3P_A^\circ + 2P_B^\circ = 3000$$

&

$$630 = P_A^\circ \left(\frac{4.5}{4.5+2+0.5}\right) + P_B^\circ \left(\frac{2}{4.5+2+0.5}\right)$$

$$4.5 P_A^\circ + 2P_B^\circ = 4410$$

$$1.5 P_A^\circ = 1410; P_A^\circ = 940 \text{ \& } P_B^\circ = 90$$

9. **Ans (C)**

$$X_{N_2} = 0.8, X_{O_2} = 0.2 \text{ in air}$$

$$\therefore P_{N_2} = 0.8 \text{ atm}, P_{O_2} = 0.2 \text{ atm}$$

$P = K_H X_{\text{gas}}$ where X is solubility of gas

$$0.8 = 2 \times 10^4 X_{N_2} \dots (1)$$

$$0.2 = 10^4 X_{O_2} \dots (2)$$

Ratio of solubility of N_2 & O_2 is 2 : 1

10. **Ans (B)**

$$r = K [A]^2 [B]^0$$

$$10^{-2} = K(1)^2(1) \Rightarrow K = 10^{-2}$$

$$\text{finally, } r = 10^{-2} (1/2)^2 (1/2)^0 = 2.5 \times 10^{-3}$$

13. **Ans (B)**

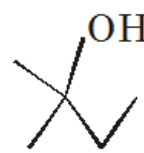
$$X \rightarrow 7/8$$

$$Y \rightarrow \frac{1}{2} \times 6$$

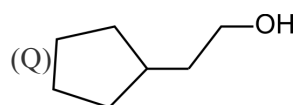
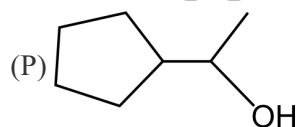
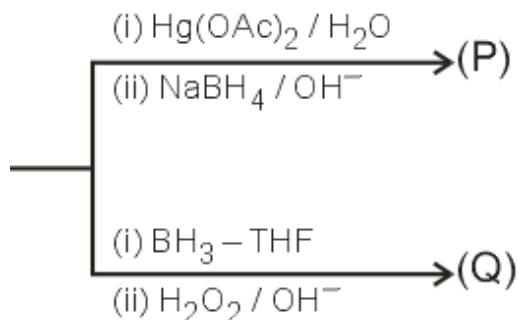
$$Z \rightarrow \frac{1}{8}$$

14. **Ans (B)**

RDS is the formation of carbocation.

Compound  gives most stable carbocation among given compounds.

19. **Ans (D)**



Both are position isomers

PART-2 : CHEMISTRY

SECTION-II

5. **Ans (64)**

For a 'n' the order reaction,

$$t_{1/2} \propto \frac{1}{a^{n-1}} \text{ where } a = \text{initial conc.}$$

$$\left(\frac{5 \times 10^{-3}}{25 \times 10^{-4}} \right)^{n-1} = \frac{8}{1}$$

$$\frac{t_{1/2}}{1} = \left(\frac{5 \times 10^{-3}}{1.25 \times 10^{-3}} \right)^{n-1}$$

$$t_{1/2} = 64 \text{ hr}$$

6. **Ans (1)**

$$\frac{y}{x} = \frac{1}{\text{antilog}(0.3)} = \frac{1}{2} = 0.50$$

7. **Ans (2)**

$$a - 2r = 1.2 \text{ \AA} \quad \text{and} \quad \sqrt{3}a = 4r$$

$$a - 2 \times \frac{\sqrt{3}}{4}a = 1.2 \text{ \AA}$$

$$a \left(1 - \frac{\sqrt{3}}{2} \right) = 1.2 \text{ \AA}$$

$$a = 8 \text{ \AA}$$

PART-3 : MATHEMATICS

SECTION-I

1. **Ans (A)**

$$\text{Here } a = 1, h = -\frac{3}{2}, b = \lambda, g = \frac{3}{2}, f = -\frac{5}{2}, c = 2$$

Angle between the lines

$$\tan^{-1} \left| \frac{2\sqrt{\frac{9}{4} - \lambda}}{\lambda + 1} \right| = \tan^{-1} \frac{1}{3}$$

$$\left| \frac{2\sqrt{\frac{9}{4} - \lambda}}{\lambda + 1} \right| = \frac{1}{3}$$

$$4 \left(\frac{9}{4} - \lambda \right) = \frac{1}{9}(\lambda + 1)^2$$

$$81 - 36\lambda = \lambda^2 + 2\lambda + 1$$

$$\lambda^2 + 38\lambda - 80 = 0$$

$$\lambda = -40, 2$$

$$\lambda = 2 \quad (\text{as } \lambda \text{ is a non negative real number})$$

2. **Ans (B)**

$$f \left(x^4 + \frac{1}{x^4} \right) = \left(x^4 + \frac{1}{x^4} \right)^3 - 3 \left(x^4 + \frac{1}{x^4} \right)$$

$$f(t) = t^3 - 3t$$

$$f'(t) = 3t^2 - 3$$

3. **Ans (A)**

$$F(x) = \int_x^{x^2} \frac{dt}{\ln t} \Rightarrow F'(x) = \frac{2x}{\ln(x^2)} - \frac{1}{\ln x} = \frac{x-1}{\ln x}$$

Thus for $x > 1$, $F(x)$ is strictly increasing.

4. **Ans (C)**

$$\text{Let } I = \int_{\pi/24}^{5\pi/24} \frac{(\cos 2x)^{1/3}}{(\cos 2x)^{1/3} + (\sin 2x)^{1/3}} dx \quad \dots(i)$$

$$\Rightarrow I = \int_{\pi/24}^{5\pi/24} \frac{(\cos \{2(\frac{\pi}{4} - x)\})^{1/3}}{(\cos \{2(\frac{\pi}{4} - x)\})^{1/3} + (\sin \{2(\frac{\pi}{4} - x)\})^{1/3}} dx$$

$$\left\{ \int_a^b f(x) dx = \int_a^b f(a+b-x) dx \right\}$$

$$\text{So } I = \int_{\pi/24}^{5\pi/24} \frac{(\sin 2x)^{1/3}}{(\sin 2x)^{1/3} + (\cos 2x)^{1/3}} dx \quad \dots(ii)$$

$$\text{Hence } 2I = \int_{\pi/24}^{5\pi/24} dx \quad [(i) + (ii)]$$

$$\Rightarrow 2I = \frac{4\pi}{24} \Rightarrow \boxed{I = \frac{\pi}{12}}$$

5. **Ans (B)**

$$I = \int_0^{\frac{\pi^2}{4}} (2 \sin \sqrt{x} + \sqrt{x} \cos x) dx$$

$$\text{Let } f(x) = 2 \sin \sqrt{x}; \quad f'(x) = \frac{\cos \sqrt{x}}{\sqrt{x}}$$

$$x f'(x) = \sqrt{x} \cos \sqrt{x}$$

$$\therefore I = \left[2x \sin \sqrt{x} \right]_0^{\frac{\pi^2}{4}} = 2 \cdot \frac{\pi^2}{4} = \frac{\pi^2}{2}$$

6. **Ans (D)**

$$\lim_{x \rightarrow 0} \frac{(\sin |x|) \cdot 2x}{3x^2} = \frac{2}{3} \frac{\sin |x|}{x}$$

7. **Ans (C)**

$$\text{LHS} = \sec x + \operatorname{cosec} x = 2\sqrt{2}$$

$$\Rightarrow x = \frac{\pi}{4} \text{ and } \frac{11\pi}{12}$$

8. Ans (D)

$$\text{Let } I = \int_{-\pi/4}^{\pi/4} \frac{2 \tan^{-1} e^x - \frac{\pi}{2}}{(1+x^2)} dx$$

$$= \int_0^{\pi/4} \frac{2 (\tan^{-1} e^x + \tan^{-1} e^{-x}) - \pi}{(1+x^2)} dx$$

(using property of definite integral)

$$= \int_0^{\pi/4} \frac{2 \cdot \frac{\pi}{2} - \pi}{(1+x^2)} dx = 0$$

9. Ans (C)

$$I = \int \frac{\sqrt[3]{\left(\frac{1+2x^3}{x^3}\right)}}{x^4} dx$$

$$= \int \frac{\sqrt[3]{\left(2 + \frac{1}{x^3}\right)^2}}{x^4} dx$$

$$\text{put } 2 + \frac{1}{x^3} = t^3$$

$$\frac{-3}{x^4} dx = 3t^2 dt$$

$$= - \int t^4 dt = \frac{-t^5}{5} = \frac{-1}{5} \left(2 + \frac{1}{x^3}\right)^{5/3} + C$$

10. Ans (A)

Take x^{15} common from Denominator

$$I = \int \frac{(1-2x)dx}{(1-x^2+x)^{\frac{1}{3}}}$$

$$\text{put } 1-x^2+x = t$$

$$\int \frac{dt}{(t)^{\frac{1}{3}}} = \frac{3}{2} t^{\frac{2}{3}} + C = \frac{3}{2} (1-x^2+x)^{\frac{2}{3}} + C$$

11. Ans (A)

$$\int \frac{x^4 dx}{x+x^5} = \int \frac{(x^4+1)}{x+x^5} - \int \frac{dx}{x+x^5}$$

$$= \int \frac{(x^4+1)dx}{x(1+x^4)} - \int \frac{dx}{x+x^5}$$

$$= \int \frac{dx}{x} - \int \frac{dx}{x+x^5}$$

$$= \log x - f(x) + c$$

12. Ans (B)

$$f'(x) = -\frac{1}{2} e^{-\frac{x}{2}} (x^2 - 8)$$

Clearly, $x = 2\sqrt{2}$ is the point of local maxima.

13. Ans (D)

$$f(x) = \begin{cases} \frac{ax^3+bx^2+cx+d}{x} & \forall x \neq 0 \\ 2 & x = 0 \end{cases}$$

Since $f(x)$ is continuous function therefore

$$\lim_{x \rightarrow 0} f(x) = 2$$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{ax^3+bx^2+cx+d}{x} = 2$$

$$\Rightarrow d = 0 \text{ \& } c = 2$$

$$\Rightarrow a = 6, b = 4, c = 2, d = 0$$

$$\Rightarrow f(x) = 6x^2 + 4x + 2$$

$$\text{therefore } p = \frac{-1}{3} \text{ and } f(p) = \frac{4}{3}$$

14. Ans (C)

$f'(x)$ must be of least degree so take

$$f'(x) = a(x-1)(x-3)$$

$$\Rightarrow f(x) = \left(\frac{a}{3}\right)x^3 - 2ax^2 + 3ax + c$$

Using $f(1) = 6$ and $f(3) = 2$

we get $a = 3, c = 2$

$$\therefore f(x) = x^3 - 6x^2 + 9x + 2$$

15. Ans (B)

$$y' = -3x^2 + 6x + 9$$

$$m = 3(-x^2 + 2x + 3)$$

$$\frac{dm}{dx} = 3(-2x + 2) = 0$$

$$\Rightarrow x = 1$$

$$\frac{d^2m}{dx^2} = -6$$

$$\Rightarrow \frac{d^2m}{dx^2} < 0$$

Hence slope (m) is maximum at

$$x = 1$$

$$\therefore m = 3(-1 + 2 + 3) = 12$$

16. Ans (D)

$$\text{Let } g(x) = f(x) - (ax^2 + bx + c).$$

Since, $g(x) = 0$ has 10 distinct real roots,

$$\Rightarrow g'(x) = 0 \text{ has minimum 9 distinct real roots,}$$

$$\Rightarrow g''(x) = 0 \text{ has minimum 8 distinct real roots,}$$

$$\Rightarrow g'''(x) = 0 \text{ has minimum 7 distinct real roots,}$$

$$\Rightarrow f'''(x) = 0 \text{ has atleast 7 distinct real roots.}$$

17. Ans (C)

$$\int_0^1 (x^2 + bx + c) dx = 1$$

$$\frac{1}{3} + \frac{b}{2} + c = 1 \Rightarrow \frac{b}{2} + c = \frac{2}{3}$$

$$3b + 6c = 4 \quad \dots(1)$$

$$P(2) = 5$$

$$4 + 2b + c = 5$$

$$2b + c = 1 \quad \dots(2)$$

from (1) & (2)

$$b = \frac{2}{9} \text{ \& } c = \frac{5}{9}$$

$$9(b + c) = 7$$

18. Ans (C)

$$I = \int_0^1 x(1-x)^{2022} dx$$

$$= \int_0^1 (1-x)x^{2022} dx$$

$$= \left[\frac{x^{2023}}{2023} - \frac{x^{2024}}{2024} \right]_0^1$$

$$= \frac{1}{2023} - \frac{1}{2024}$$

19. Ans (D)

Let tangent at (t^2, t^3) passes through $\left(0, -\frac{1}{2}\right)$

$$2y \frac{dy}{dx} = 3x^2 \Rightarrow \frac{dy}{dx} = \frac{3x^2}{2y} = \frac{3t}{2}$$

$$\frac{dy}{dx} = \frac{y_2 - y_1}{x_2 - x_1} \Rightarrow \frac{t^3 + \frac{1}{2}}{t^2} = \frac{3t}{2}$$

$$\Rightarrow 2t^3 + 1 = 3t^3 \Rightarrow t = 1$$

$$\Rightarrow \text{Equation of tangent is } y - 1 = \frac{3}{2}(x - 1)$$

$$\Rightarrow 2y - 2 = 3x - 3 \Rightarrow 3x = 2y + 1$$

20. Ans (A)

Note that f is bijective hence f^{-1} exist

when $y = 4$ $(f^{-1}(4) = x \Rightarrow f(x) = 4)$

$$2x^3 + 7x - 9 = 0$$

$$2x^2(x-1) + 2x(x-1) + 9(x-1) = 0$$

$$(x-1)(2x^2 + 2x + 9) = 0$$

$x = 1$ only

$\Rightarrow A$; as $2x^2 + 2x + 9 = 0$ has no real roots

PART-3 : MATHEMATICS

SECTION-II

1. Ans (5)

$$\text{Let } I = \int_6^{16} \frac{\log_e x^2}{\log_e x^2 + \log_e (x^2 - 44x + 484)} dx$$

$$I = \int_6^{16} \frac{\log_e x^2}{\log_e x^2 + \log_e (x - 22)^2} dx \dots(1)$$

we know

$$\int_a^b f(x) dx = \int_a^b f(a+b-x) dx \text{ (king)}$$

$$\text{So } I = \int_6^{16} \frac{\log_e (22-x)^2}{\log_e (22-x)^2 + \log_e (22-(22-x))^2} dx$$

$$(1) + (2)$$

$$2I = \int_6^{16} 1 dx = 10$$

$$I = 5$$

2. Ans (0)

$$I = \int_0^1 0 dx = 0$$

3. Ans (1)

$$I = \left(e^x \left((x-1)^3 - 3(x-1)^2 + 6(x-1) - 6 \right) \right)_0^1$$

$$= -6e - (-1 - 3 - 6 - 6) = 16 - 6e$$

$$\Rightarrow [I] = -1$$

$$\therefore |[I]| = 1$$

4. **Ans (15)**

$$I = \int_0^{7\pi} |\cos x| dx + \int_0^{2\pi/5} |\cos x| dx$$

$$= 7 \int_0^{\pi} |\cos x| dx + \int_0^{\pi/2} \cos x dx - \int_{\pi/2}^{2\pi/3} \cos x dx$$

$$= 14 + 1 + \left(1 - \frac{\sqrt{3}}{2}\right)$$

$$I = 16 - \frac{\sqrt{3}}{2}$$

$$[I] = 15$$

5. **Ans (1)**

Apply L'Hopital rule

$$\lim_{x \rightarrow 0} \frac{\sec^3(x^4) \cdot 2x}{x \cos x + \sin x}$$

$$\lim_{x \rightarrow 0} \frac{\sec^3(x^4) \cdot 2}{\cos x + \frac{\sin x}{x}} \Rightarrow \frac{2 \times 1}{1 + 1} \Rightarrow 1$$

6. **Ans (6)**

$$\int_a^{\beta} f(x) dx + \int_a^b f^{-1}(x) dx = b\beta - a\alpha$$

(if $f(a) = a, f(\beta) = b$)

$$\therefore I_1 + I_2 = 2 \times 3 - 0 \times 1 = 6$$

7. **Ans (4)**

Let $x + 4 = y \Rightarrow dx = dy$

$$I = \int e^{y-4} \left(\frac{y-2}{y}\right)^2 dy = e^{-4} \int e^y \left(\frac{y^2 - 4y + 4}{y^2}\right) dy$$

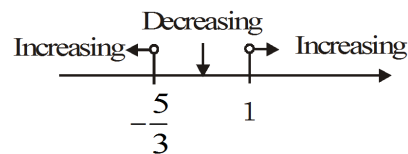
$$= e^{y-4} \left(1 - \frac{4}{y}\right) + C = \frac{x}{x+4} e^x + C$$

$$\Rightarrow \alpha = 0 \text{ \& } \beta = 4$$

8. **Ans (3)**

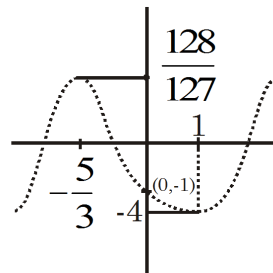
$$f(x) = x^3 + x^2 - 5x - 1$$

$$f'(x) = 3x^2 + 2x - 5 = (x-1)(3x+5)$$



$$x \rightarrow \infty : f(x) \rightarrow \infty$$

$$x \rightarrow -\infty : f(x) \rightarrow -\infty$$



$$\therefore f(1) = -4$$

$$f\left(-\frac{5}{3}\right) = \frac{128}{127};$$

$$f(10) = -1 < 0$$

$$\left. \begin{array}{l} f(1) = -4 < 0 \\ f(2) = 1 > 0 \end{array} \right\} \text{one root lies between 1 \& 2}$$

$$\left. \begin{array}{l} f(-1) = 4 > 0 \\ f(0) = -1 < 0 \end{array} \right\} \text{one root lies between 0 \& -1}$$

$$\left. \begin{array}{l} f(-2) = 5 > 0 \\ f(-3) = -4 < 0 \end{array} \right\} \text{one root lies between -2 \& -3}$$

$$\therefore [\alpha] + [\beta] + [\gamma]$$

$$= [\text{value between 1 \& 2}] + [\text{value between 0 \& -1}]$$

$$+ [\text{value between -2 \& -3}]$$

$$= 1 - 1 - 3 = -3$$

9. **Ans (5050)**

$$f(x) = \prod_{r=1}^{100} (x-r)^{r(101-r)}$$

$$\ln(f(x)) = \sum_{r=1}^{100} r(101-r) \ln(x-r)$$

$$\Rightarrow \frac{f'(x)}{f(x)} = \sum_{r=1}^{100} \frac{r(101-r)}{(x-r)}$$

$$\therefore \frac{f'(101)}{f(101)} = \sum_{r=1}^{100} \frac{r(101-r)}{(101-r)} = \sum_{r=1}^{100} r$$

$$\therefore P = 50 \times 101$$

10. Ans (2)

$$\int \frac{xe^x + e^x}{(1 + xe^x)^2} dx$$

put $xe^x = t$

$$\Rightarrow e^x(x+1)dx = dt$$

$$\therefore I = \int \frac{dt}{(1+t)^2} = -\frac{1}{(1+t)} + \lambda$$

$$= -\frac{1}{1+xe^x} + \lambda = f(x)$$

where λ is constant of integration

$$\therefore f(0) = 1 \Rightarrow \lambda = 2$$

$$\therefore f(x) = \frac{2xe^x + 1}{xe^x + 1}$$

$$\therefore \lim_{x \rightarrow \infty} f(x) = 2$$